R^2 Flow: Recursive Self-Improvement via Recursive Skill Evolution

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HarnessRSI 69 ★★

evidence: retained

Paper

Code

1 What changes, what is kept

What changes: Supervisor weights + skill library
executor stays frozen (Qwen3.5-9B)

1. **Train one phase**
   library S(k) fixed; flow objective R2 TB on a shared-state graph (reorderings merge)
2. **Read two scores per skill**
   flow share (how much flow) and signed utility (does it help?), kept separate
3. **Whether: verifier evidence**
   independent verifiers label calls; wide interval = defer; upper bound low = refine / prune
4. **Paired validation**
   held-out queries: non-inferior on success, reward, cost, latency, else roll back
5. **Commit S(k+1), next phase**
   when residual variance plateaus; mean OOD score over 8 phases: 71.2 → 81.02

2 Headline vs strongest baseline

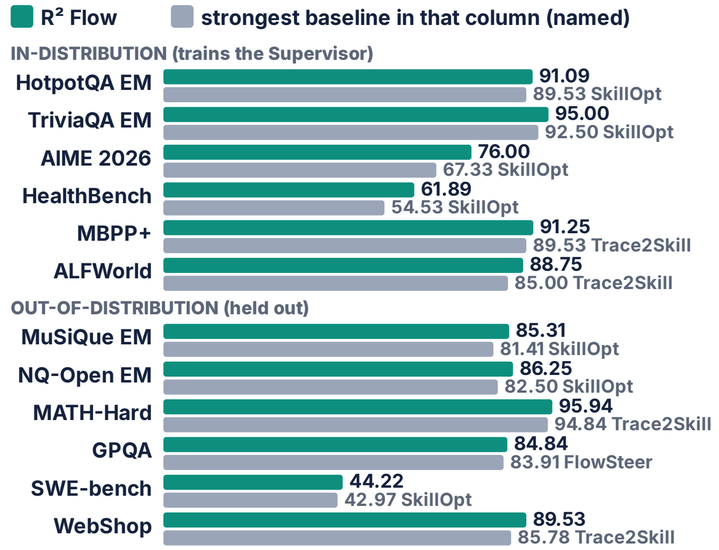
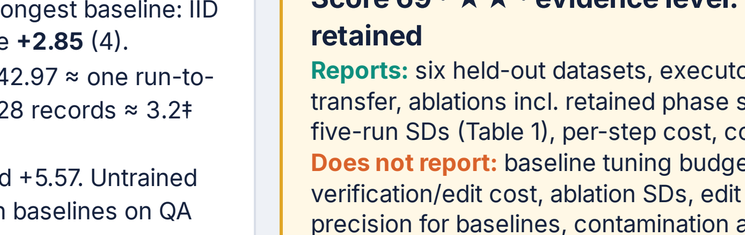


Table 1, primary metric, five-run mean, one frozen Qwen3.5-9B executor shared by all methods. 128 records per dataset (AIME: 30).

3 Controls that decide if the gain is real

Ablation · IID points below full (Table 2, all 9 variants, no SD)



All 108 ablated cells are below full (0.31 to 14.00).

OOD, Fig 5c–e	score	edits (Fig 5d)
R^2 Flow (full)	71.2 → 81.02	33/37 raise verified
Fixed phase	78.99	–
Reward labels	peak 77.9, falls	25/44: reward ↑ verified ↓
Frozen library	73.4–74.8	–

Frozen library: scores after phase 1. Harmful skills kept (of 12 injected): R^2 Flow 1 · share-only 8 · Bayesian-Agent 6 · SkillFlow 7.

R^2 TB vs GRPO / SubTB, same framework: +7.15 / +2.25 IID; +7.5% whole-step GPU vs GRPO. Six other executors: +15.70 to +24.64 vs direct.

Takeaway

Verifier-gated library edits plus a shared-state flow objective beat RL and skill-evolution baselines at one shared executor by 2 to 6 points on the averages. Ablations favour each component, but several cells differ by about one SD, part of the gain over the backbone exists before training, and baseline budgets and total costs are not described.

4 Where the gain is thin, unverified or not stated

- Table 1 Δ (+33.1 … +63.0) is vs the direct-answer backbone (near the bottom). Vs the strongest baseline: IID EM avg **+2.03** (2 sets), accuracy-type **+5.57** (4 sets); OOD EM **+3.83** (2), accuracy-type **+2.85** (4).
- **Narrow cells:** GPQA 84.84 vs 83.91, MATH-Hard 95.94 vs 94.84, SWE-bench 44.22 vs 42.97 ≈ one run-to-run SD (1.18 / 0.86 / 1.18); some baseline SDs are 0.00. Binomial SE of one accuracy on 128 records ≈ 3.2‡ pts.
- **AIME** (30 problems, 3.33 pts each ‡): 76.00 vs 67.33 is 39%‡ of the summed gain behind +5.57. Untrained architecture: +13.71 of +31.78‡ IID points over the backbone, yet far below skill-evolution baselines on QA (HotpotQA 77.03 vs 88.12–89.53).
- **Edit precision** 0.89 only vs the "− Verifier" ablation (0.43), not vs Trace2Skill / SkillOpt. IID evaluation records not stated disjoint from IID training queries.
- **Not stated:** baseline tuning budgets, verifier access; verification/edit cost; what varies across runs; ablation SDs. FlowSteer, SkillFlow share the first author. No limitations section.

Our read · HarnessRSI

Score 69 · ★★ · evidence level: retained

Reports: six held-out datasets, executor transfer, ablations incl. retained phase state, five-run SDs (Table 1), per-step cost, code.
Does not report: baseline tuning budgets, verification/edit cost, ablation SDs, edit precision for baselines, contamination audit.
Judgment notes: variance marked unclear, but Table 1 has SDs; leakage_guard rests on a harmful-skill test.